a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?
b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.
c. Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$. Using these numerical values, show that $\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.
d. Use the least squares estimates from part (b) to compute the fitted values of y, and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y, $s_y^2 = \sum_{i=1}^{N}(y_i-\bar{y})^2/(N-1)$, the sample variance of x, $s_x^2 = \sum_{i=1}^{N}(x_i-\bar{x})^2/(N-1)$, the sample covariance between x and y, $s_{xy} = \sum_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})/(N-1)$, the sample correlation between x and y, $r_{xy} = s_{xy}/(s_x s_y)$ and the coefficient of variation of x, $CV_x = 100(s_x/\bar{x})$. What is the median, 50th percentile, of x?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4				
2	2				
1	3				
−1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$	$\sum \hat{e}_i =$	$\sum \hat{e}_i^2 =$	$\sum x_i\hat{e}_i =$

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.
f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.
g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.
h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i/N$.
i. Compute $\hat{\sigma}^2$.
j. Compute $\widehat{\text{var}}(b_2|\mathbf{x})$ and $\text{se}(b_2)$.

a. $\sum x_i = 3+2+1-1+0 = 5$

$\sum y_i = 4+2+3+1+0 = 10$

$\bar{x} = \frac{5}{5} = 1 \qquad \bar{y} = \frac{10}{5} = 2$

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
−1	1	−2	4	−1	2
0	0	−1	1	−2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum(x_i-\bar{x}) = 0$	$\sum(x_i-\bar{x})^2 = 10$	$\sum(y_i-\bar{y}) = 0$	$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8$

b. $b_2 = \dfrac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} = \dfrac{8}{10} = 0.8$

$b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8(1) = 1.2$

b_2 (slope) $= 0.8$ when X increase 1 unit, the expected average value of y increase 0.8 unit.

b_1 (intercept) $= 1.2$ when $X=0$, the expected average value of y is 1.2.

c. x_i^2 : 9, 4, 1, 1, 0 $\Rightarrow \sum_{i=1}^{5} x_i^2 = 15$

$x_i y_i$: 12, 4, 3, −1, 0 $\Rightarrow \sum_{i=1}^{5} x_i y_i = 18$

$\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$

$10 = 15 - 5(1^2) = 10$

左右相等，得證

$\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$

$8 = 18 - 5(1)(2) = 8$

左右相等，得證 ※

d. $\hat{y} = 1.2 + 0.8x$

$\hat{e} = y - \hat{y}$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	$1.2+0.8(3)=3.6$	$4-3.6=0.4$	0.16	$3(0.4)=1.2$
2	2	$1.2+1.6=2.8$	$2-2.8=-0.8$	0.64	$2(-0.8)=-1.6$
1	3	$1.2+0.8=2$	$3-2=1$	1	$1(1)=1$
-1	1	$1.2-0.8=0.4$	$1-0.4=0.6$	0.36	$-1(0.6)=-0.6$
0	0	$1.2+0=1.2$	$0-1.2=-1.2$	1.44	$0(-1.2)=0$
$\sum x_i=5$	$\sum y_i=10$	$\sum \hat{y}_i=10$	$\sum \hat{e}_i=0$	$\sum \hat{e}_i^2=3.6$	$\sum x_i\hat{e}_i=0$

$$\Sigma (y_i-\bar{y})^2 = 2^2 + 0 + 1^2 + (-1)^2 + (-2)^2 = 10$$

$$s_y^2 = \frac{\Sigma (y_i-\bar{y})^2}{N-1} = \frac{10}{4} = 2.5$$

$$s_x^2 = \frac{\Sigma (x_i-\bar{x})^2}{N-1} = \frac{10}{4} = 2.5$$

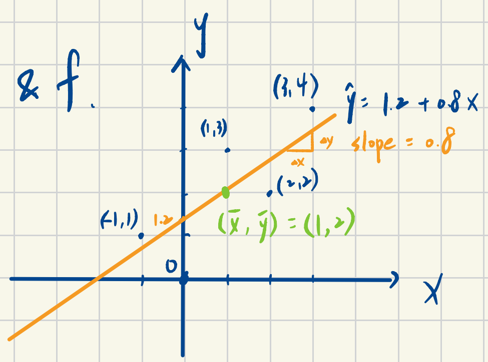
$$s_{xy} = \frac{\Sigma (x_i-\bar{x})(y_i-\bar{y})}{N-1} = \frac{8}{4} = 2$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2.5}\sqrt{2.5}} = \frac{2}{2.5} = 0.8$$

$$CV_x = 100 \times \left(\frac{s_x}{\bar{x}}\right) = 100 \times \left(\frac{\sqrt{2.5}}{1}\right) \approx 100 \times 1.5811 = 158.11$$

Median of $x \Rightarrow (-1, 0, 1, 2, 3)$ $\therefore$ median of x is 1.

e. & f.



$\because b_1 = \bar{y} - b_2\bar{x}$

$\therefore$ the regression line indeed pass through the point $(\bar{x}, \bar{y}) = (1,2)$ #

g. $\bar{y} = b_1 + b_2 \bar{x}$

$2 = 1.2 + 0.8(1)$ 左右相等,得證 ※

h. $\sum \hat{y}_i = 10$

$\bar{\hat{y}} = \dfrac{\sum \hat{y}_i}{N} = \dfrac{10}{5} = 2 = \bar{y}$

$\therefore \bar{y} = \bar{\hat{y}}$ ※

i. $\hat{\sigma}^2 = \dfrac{\sum \hat{e}_i^2}{N-2} = \dfrac{3.6}{5-2} = \dfrac{3.6}{3} = 1.2$ ※

j. $\widehat{var}(b_2|x) = \dfrac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \dfrac{1.2}{10} = 0.12$ ※

$se(b_2) = \sqrt{0.12} \approx 0.3464$ ※